Marco Soprani Tayar

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EDUCATION

University of São Paulo (EESC/USP) · B.Eng. Mechatronics Engineering

2023 - December 2027 (expected)

RESEARCH & ENGINEERING EXPERIENCE

Researcher · USP Center for Robotics (CROB)

Mobile Robotics Group · March 2024 - present · Advisor: Prof. Marcelo Becker

- **Petrobras / Embrapii:** developed visual relocalization using live-camera matching against a digital twin reconstructed from monocular video with 3D Gaussian splatting, for odometry correction and robot-position updates in ground control.
- **Volkswagen:** developing warehouse coverage planning using lidar-inertial odometry and map information; previously worked on offboard vision-based box identification for UAV inventory.
- **VLN on the Fly:** deployed a quantized vision-language model onboard and integrated its image targets with depth to produce spatial navigation goals; deployed VLA baselines onboard and in simulation.
- Quantized and deployed vision networks for onboard inference; built simulation environments, designed rewards, and trained and compared RL navigation policies.

Research Intern · Texas A&M University

Advanced Vertical Flight Laboratory · January - March 2026 · Advisor: Dr. Moble Benedict

- Built the autonomous flight stack, implemented navigation nodes and visual-inertial odometry, and modified ArduPilot firmware to switch EKF estimation sources between flight modes.
- Deployed and field-tested the system, successfully demonstrating in-flight transitions between GPS and vision-based state estimation.

Avionics Intern · Embraer Defense

January - March 2025

- Worked on avionics systems and sensor integration for military aircraft.

Aerial Robotics Director · SEMEAR Robotics Group, EESC/USP

Director: November 2024 - December 2025 · Member: May 2023 - December 2025

- Directly led a 40-person team building an autonomous indoor drone; placed 4th in the IMAV 2025 indoor competition and received the Best Indoor Design Award for the team's system design.
- Developed autonomous flight software in Python and ROS, integrating computer vision for international competitions.

PUBLICATIONS · FIRST AUTHOR

M. S. Tayar et al. VLN on the Fly: An Onboard Vision-Language Navigation Stack for Aerial Robots. IMAV 2026.

M. S. Tayar et al. Autonomous UAV Flight Navigation in Confined Spaces: A Reinforcement Learning Approach. LARS 2025, IEEE.

ADDITIONAL AWARDS & SERVICE

IMAV team awards: Most Promising Team, 2026 (4th indoor); Autonomy Prize, 2024 (3rd indoor). **AWS DeepRacer:** national finalist, CSBC 2024. **Reviewer:** IMAV 2026.

TECHNICAL SKILLS & LANGUAGES

Learning & perception: reinforcement learning, sim-to-real, VLM/VLA deployment, model quantization, visual relocalization, VIO/LIO, 3D Gaussian splatting.

Software & robotics: Python, C++, PyTorch, ROS/ROS 2, ArduPilot, PX4, Gazebo, Isaac Sim, ONNX, TensorRT, Docker, Linux, Git.

Languages: Portuguese (native); English (C2, EF SET 89/100); Spanish (professional); Mandarin (beginner).